SUPERCRITICAL VARIATIONAL CURVATURE BARRIERS FOR RICCI-TYPE FLOWS ON COMPACT S^3

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2020 *Mathematics Subject Classification.* 53E20, 53C44, 58J35.
Key words and phrases. Ricci-type flow, curvature barrier, Berger spheres, Perelman entropy, noncollapse.

Abstract. This paper is not a revision of the operational Planck-boundary interpretation used in the compact S^3 programme. In that setting, ℓ_P marks the limit of physical distinguishability, not a dynamical forcing term in the Ricci flow. The present note asks a separate mathematical question: if Ricci-type flows are treated as autonomous geometric dynamics on compact S^3, which variational curvature barriers can suppress supercritical curvature growth before a prescribed curvature scale is reached?

A Hopf–Berger analysis of Ricci-type flows on compact S^3 shows that bounded traceless forcing, fibre-restoring forcing, and a constant cosmological-constant term do not supply the negative cubic damping needed to control the round-collapse direction. The homogeneous term $h = 2\lambda|\mathrm{Rm}|g$ has the desired sign in a finite-dimensional reduction, but fails as a PDE mechanism because the discarded Laplacian source $-4\lambda\Delta|\mathrm{Rm}|$ is of the same scale as the intended damping near a curvature concentration.

This paper isolates the corresponding variational replacement. The local high-curvature regulator is the negative L^2-gradient of a curvature energy

$$\mathcal{F}_\Phi[g] = \int_{S^3} \Phi(K)\, d\mu_g, \qquad K = |\mathrm{Rm}|^2,$$

and, in the singular wall version,

$$\mathcal{B}_{p,q}[g] = \int_{S^3} \frac{(K+\delta)^q}{(K_P - K)^p}\, d\mu_g, \qquad K_P = \ell_P^{-4}.$$

The algebraic high-curvature test is passed. In dimension three

$$K = |\mathrm{Rm}|^2 = 4|\mathrm{Ric}_0|^2 + \frac{1}{3}R^2,$$

so K detects both round and anisotropic curvature, and the conformal variation gives the decisive factor

$$4K\Phi'(K) - 3\Phi(K).$$

For $\Phi(K) = K^q$ this is $(4q-3)K^q$, so the sharp scale threshold in dimension three is $q > 3/4$. For the Planck wall the same factor diverges with the correct sign as $K \nearrow K_P$. On the full Berger family the wall has an exact two-dimensional reduction and the anisotropy ratio $u = z/x$ satisfies

$$\dot{u} = \frac{8u(1-u)}{x}\left(1 + \frac{2\varepsilon\Phi'(K)}{x}(11u-4)\right),$$

which yields an invariant cone $u \geq 4/11$ and identifies the lower Hopf sector where the wall may accelerate fibre collapse. The same reduction gives closed formulas for volume, scalar curvature, traceless Ricci size, and the signs of the Planck-wall Euler tensor on Berger sectors. A simple zero-mode heat-kernel lower bound also shows that the scale detector $\sup_{0<t\leq r^2} t^{3/2} H_g(t,x,x)$ diverges along fixed-base Hopf collapse. A wall gap gives an explicit Ricci lower bound and hence controlled heat-kernel comparison constants, although no propagation theorem for this detector is claimed.

The paper also records the exact obstruction to turning this wall into a noncollapse theorem by an additive Perelman-potential repair. On the homogeneous Berger sector the candidate $\lambda_\Phi = \lambda_1(-4\Delta + R + \varepsilon\Phi(K))$ reduces exactly to $R + \varepsilon\Phi(K)$. For $\Phi(K) = K$, at the round point,

$$\dot{\lambda}_\Phi = \frac{12(2x-\varepsilon)(x+4\varepsilon)}{x^4},$$

so monotonicity fails for every fixed $\varepsilon > 0$ once $x < \varepsilon/2$. More generally, additive matching K-potential repairs of the form

$$\mathcal{W}_{\mathrm{Perelman}} + \sigma\varepsilon\tau^\alpha K$$

fail the same Berger test for every sign σ and every exponent α. For the singular Planck wall $\Phi(K) = (K+\delta)^q(K_P - K)^{-p}$, the corresponding additive repair also fails analytically at the wall: its Berger derivative has a negative leading term of order

$$-\varepsilon^2(K_P - K)^{-2p-2}.$$

Thus the curvature wall is a valid local high-curvature metric regulator, but it is not itself a noncollapse entropy. By contrast, the unmodified Perelman entropy passes the protected positive-scalar Berger cone test: on $4/11 \leq u < 4$ its reduced derivative along the wall flow is nonnegative. This gives the first positive non-additive two-mechanism result in the paper. A complete internal replacement for surgery would still require a corresponding propagation or monotonicity theorem for the unmodified entropy along the full inhomogeneous DeTurck-gauged wall flow.

1. Purpose

The status of this paper is deliberately mathematical. It does not claim that the operational Planck-boundary interpretation requires a curvature-wall correction, nor does it replace that interpretation. In the compact S^3 Planck-boundary programme, ℓ_P marks a limit of physical distinguishability. Here we ask a different question: once modified Ricci-type flows are studied as autonomous geometric systems, can one construct intrinsic variational forcing terms that prevent supercritical curvature growth before a prescribed curvature scale is reached?

The object is a Ricci-type flow on a compact closed three-sphere

$$\partial_\tau g = -2\mathrm{Ric} + h[g].$$

The problem is not to imitate Ricci-flow surgery. Surgery is an external topological intervention. The question here is narrower: can an internal, geometrically natural forcing term prevent a high-curvature collapse inside a fixed closed slice?

The useful test is the curvature reaction estimate

$$(\partial_\tau - \Delta)|\mathrm{Rm}|^2 = -2|\nabla \mathrm{Rm}|^2 + C_{\mathrm{RF}}(\mathrm{Rm}), \qquad |C_{\mathrm{RF}}(\mathrm{Rm})| \le C_3|\mathrm{Rm}|^3.$$

A local forcing can dominate the singular regime only if, at large curvature, its curvature contribution contains a negative term of order strictly stronger than, or at least cubic with coefficient exceeding, the Ricci reaction:

$$2\langle C_h(\mathrm{Rm}), \mathrm{Rm}\rangle \le -\gamma|\mathrm{Rm}|^3 + A|\mathrm{Rm}|^2 + B, \qquad \gamma > C_3.$$

A constant cosmological term does not pass this test. If $h = 2\Lambda g$, then its contribution is

$$-4\Lambda|\mathrm{Rm}|^2,$$

which is quadratic. It cannot dominate $C_3|\mathrm{Rm}|^3$ as $|\mathrm{Rm}| \to \infty$.

The preceding Hopf–Berger calculation also gives a geometric warning. For

$$g_{a,c} = a^2(\sigma_1^2 + \sigma_2^2) + c^2\sigma_3^2 \quad \text{on } SU(2) \simeq S^3,$$

one has

$$|\mathrm{Rm}|^2 = \frac{4(16a^4 - 24a^2c^2 + 11c^4)}{a^8}.$$

Equivalently, with $x = a^2$, $z = c^2$, and $u = z/x$,

$$K := |\mathrm{Rm}|^2 = \frac{4(16 - 24u + 11u^2)}{x^2}.$$

Thus $c \to 0$ at fixed a is a bounded-curvature fibre collapse, while $a \to 0$ at fixed u is the genuine curvature singularity. Any universal regularizer must see the round/base collapse. Traceless-only or fibre-only invariants do not suffice.

2. Operational meaning of universality

The word "universal" must be made finite. We use it in the following operational sense.

Definition 1 (local high-curvature regularizer)**.** *A forcing $h[g]$ is called a local high-curvature regularizer on compact three-manifolds if it satisfies four tests.*

- *(i) It is natural: $h[\phi^* g] = \phi^* h[g]$ for diffeomorphisms ϕ.*
- *(ii) It is local: it is built from g, curvature, and finitely many covariant derivatives.*
- *(iii) Its scalar detector does not vanish on a non-flat three-dimensional curvature tensor.*
- *(iv) In a high-curvature concentration it supplies negative leading-order curvature drift strong enough to dominate $C_3|\mathrm{Rm}|^3$, without discarding spatial derivative terms of the same order.*

This definition deliberately excludes a common false success. A term such as $2f(|\mathrm{Rm}|)g$ may look good in a homogeneous reduction, but its scalar curvature variation includes Laplacians of $f(|\mathrm{Rm}|)$. For $f = \lambda|\mathrm{Rm}|$,

$$\delta R = -2fR - 4\Delta f$$

in dimension three. The homogeneous model keeps $-2\lambda|\mathrm{Rm}|R$ and drops $-4\lambda\Delta|\mathrm{Rm}|$. Near a neck concentration of width L and peak $|\mathrm{Rm}|_{\mathrm{pk}} \sim L^{-2}$, the dropped term scales like

$$\lambda\frac{|\mathrm{Rm}|_{\mathrm{pk}}}{L^2} \sim \lambda|\mathrm{Rm}|^2_{\mathrm{pk}},$$

the same order as the intended damping. It is therefore not a stable PDE mechanism.

The replacement is to keep the derivative terms by making the forcing variational.

3. The detector: $K = |\mathrm{Rm}|^2$

In dimension three the Weyl tensor vanishes, and the full curvature tensor is determined by Ric. The norm identity is

$$|\mathrm{Rm}|^2 = 4|\mathrm{Ric}|^2 - R^2 = 4|\mathrm{Ric}_0|^2 + \frac{1}{3}R^2.$$

Here

$$\mathrm{Ric}_0 = \mathrm{Ric} - \frac{1}{3}Rg$$

is the traceless Ricci tensor.

Lemma 1 (minimal scalar detector)**.** *In dimension three, $K = |\mathrm{Rm}|^2$ is the minimal positive scalar curvature detector for the present problem: it detects both the round direction* ($\mathrm{Ric}_0 = 0$, $R \neq 0$) *and anisotropic curvature* ($\mathrm{Ric}_0 \neq 0$)*. A traceless-only detector misses the round collapse; scalar curvature alone does not give a positive definite norm of the whole curvature tensor; derivative detectors such as $|\nabla\mathrm{Rm}|^2$ miss homogeneous round collapse.*

Proof. The identity above is pointwise. The term $R^2/3$ is precisely the round component, while $4|\mathrm{Ric}_0|^2$ is the anisotropic component. Removing either term loses one of the two curvature channels. A derivative-only detector vanishes on homogeneous Berger metrics, including the round shrinking mode, even while $K \to \infty$. □

Thus the basic invariant must be a function of

$$K = |\mathrm{Rm}|^2.$$

4. Variational forcing

Let

$$\mathcal{F}_\Phi[g] = \int_M \Phi(K)\, d\mu_g, \qquad K = |\mathrm{Rm}|^2,$$

where Φ is smooth on the relevant interval. Define $E_\Phi(g)$ by

$$D\mathcal{F}_\Phi[g](v) = \int_M \langle E_\Phi(g), v\rangle_g \, d\mu_g.$$

The proposed forced flow is

$$\boxed{\partial_\tau g = -2\mathrm{Ric} - \varepsilon E_\Phi(g)}$$

with $\varepsilon > 0$.

This is a fourth-order geometric flow, because K depends on two derivatives of the metric and E_Φ contains the Euler–Lagrange terms coming from varying curvature. That is not a defect. The defect of $f(K)g$ was precisely that the fourth-order derivative structure was hidden and then lost in the homogeneous reduction.

Proposition 1 (scale derivative)**.** *Let M have dimension $n = 3$. Under the constant conformal scaling $g_s = e^{2s}g$,*

$$K_s = e^{-4s}K, \qquad d\mu_{g_s} = e^{3s}d\mu_g.$$

Hence

$$\left.\frac{d}{ds}\right|_{s=0} \mathcal{F}_\Phi[g_s] = \int_M \big(3\Phi(K) - 4K\Phi'(K)\big)\, d\mu_g.$$

Consequently the negative L^2-gradient has the correct high-curvature conformal sign whenever

$$4K\Phi'(K) - 3\Phi(K) > 0.$$

Proof. Substitution gives

$$\mathcal{F}_\Phi[g_s] = \int_M e^{3s}\Phi(e^{-4s}K)d\mu_g.$$

Differentiating at $s = 0$ gives the displayed identity. If E_Φ is the L^2-gradient and $v = 2g$, then

$$\int_M \langle E_\Phi, 2g\rangle \, d\mu = \int_M (3\Phi - 4K\Phi') \, d\mu.$$

Thus the negative gradient expands the metric in the high-curvature conformal mode precisely when $4K\Phi' - 3\Phi$ is positive. □

5. Power-law regularizers

Take

$$\Phi(K) = K^q.$$

Then

$$4K\Phi'(K) - 3\Phi(K) = (4q - 3)K^q.$$

Thus the conformal sign is correct exactly when

$$q > \frac{3}{4}.$$

The same exponent is the scale threshold. Under a length scale L,

$$|\mathrm{Rm}| \sim L^{-2}, \qquad K = |\mathrm{Rm}|^2 \sim L^{-4}, \qquad d\mu \sim L^3,$$

and therefore

$$\int K^q \, d\mu \sim L^{3-4q}.$$

The functional is scale-critical at $q = 3/4$ and blows up on shrinking curvature concentrations when $q > 3/4$.

Remark 1 (correction of the exponent)**.** *The exponent $q > 3/2$ is stronger than necessary for the scale and conformal tests in dimension three. The critical exponent for $\int |\mathrm{Rm}|^{2q} d\mu = \int K^q d\mu$ is $q = 3/4$. The condition $q > 3/2$ may be retained as a deliberately aggressive, non-minimal choice, but it should not be called the minimal scale threshold.*

In a round high-curvature mode the contribution to K has the leading form

$$\partial_\tau K\big|_{\mathrm{var}} = -\frac{\varepsilon}{3}(4q - 3)K^{q+1} + \text{derivative and lower-order terms.}$$

The Ricci-flow reaction has order $K^{3/2}$. Thus every

$$q > \frac{1}{2}$$

beats the cubic Ricci reaction by degree, and every $q > 3/4$ both beats it and passes the scale/conformal test.

The power-law candidate is therefore

$$\boxed{\partial_\tau g = -2\mathrm{Ric} - \varepsilon \, \mathrm{grad}_{L^2} \int_{S^3} (|\mathrm{Rm}|^2 + \delta)^q \, d\mu_g, \qquad q > \frac{3}{4}, \quad \delta > 0.}$$

The parameter $\delta > 0$ makes the integrand smooth at $K = 0$ when q is not an integer.

6. Planck barrier

The power-law energy punishes concentration, but it does not by itself encode a finite ceiling. A genuine curvature wall is obtained by restricting to

$$\mathcal{U}_P = \{g : \sup_M K < K_P\}, \qquad K_P = \ell_P^{-4},$$

and using

$$\mathcal{B}_{p,q}[g] = \int_M \frac{(K+\delta)^q}{(K_P - K)^p}\, d\mu_g, \qquad p > 0, \quad q > \frac{3}{4}, \quad \delta > 0.$$

The proposed Planck-barrier flow is

$$\boxed{\partial_\tau g = -2\mathrm{Ric} - \varepsilon\ \mathrm{grad}_{L^2}\, \mathcal{B}_{p,q}[g].}$$

Proposition 2 (barrier sign)**.** *For*

$$\Phi(K) = \frac{(K+\delta)^q}{(K_P - K)^p}, \qquad 0 \le K < K_P,$$

one has

$$4K\Phi'(K) - 3\Phi(K) = \Phi(K)\left(\frac{4qK}{K+\delta} + \frac{4pK}{K_P - K} - 3\right).$$

In particular, for large K and especially as $K \nearrow K_P$, the negative gradient gives an inward high-curvature drift, with leading blow-up

$$4K\Phi'(K) - 3\Phi(K) \sim \frac{4pK(K+\delta)^q}{(K_P - K)^{p+1}}.$$

Consequently the conformal contribution to K has leading sign

$$\partial_\tau K\big|_{\mathrm{bar}} \sim -\frac{4\varepsilon p}{3}\frac{K^2(K+\delta)^q}{(K_P - K)^{p+1}} \quad \text{as } K \nearrow K_P.$$

Proof. Differentiate logarithmically:

$$\frac{\Phi'(K)}{\Phi(K)} = \frac{q}{K+\delta} + \frac{p}{K_P - K}.$$

Multiplication by $4K$ and subtraction of 3 give the formula. The final asymptotic follows from the pole at $K = K_P$. The factor $K/3$ in the drift of K is the conformal scaling relation in dimension three. □

Remark 2 (choice of p)**.** *For the pointwise wall, $p > 0$ already creates a singular barrier. Taking $p > 1$ or $p > 3/2$ is a stronger modelling choice. It may be useful when one wants the variational force to become extremely stiff before K reaches K_P, but it is not required by the elementary scale test.*

7. Berger verification

For the Berger family,

$$K = \frac{4(16 - 24u + 11u^2)}{x^2}, \qquad x = a^2, \quad u = \frac{c^2}{a^2}.$$

The round point is $u = 1$, where

$$K = \frac{12}{x^2}.$$

The unforced Berger calculation gives

$$\left.\frac{\partial_\tau K}{K^{3/2}}\right|_{u=1} = \frac{4}{\sqrt{3}}.$$

Thus the Ricci reaction is positive cubic in $|\mathrm{Rm}|$, equivalently positive $K^{3/2}$.

For the power-law variational forcing, the leading conformal high-curvature term is

$$-\frac{\varepsilon}{3}(4q-3)K^{q+1}.$$

For $q > 3/4$,

$$K^{q+1} \gg K^{3/2} \qquad (K \to \infty),$$

so the proposed forcing passes the Berger round-collapse degree test.

For the Planck barrier,

$$\partial_\tau K\big|_{\mathrm{bar}} \sim -\frac{4\varepsilon p}{3}\frac{K^2(K+\delta)^q}{(K_P-K)^{p+1}}, \qquad K \nearrow K_P,$$

so the leading formal drift points away from the forbidden boundary $K = K_P$.

This calculation also explains why the Hopf fibre is not the missing regularizing channel. At $c \to 0$ with a fixed,

$$K \to \frac{64}{a^4} < \infty.$$

The singular variable is a, not c. The invariant K sees this without imposing a preferred fibre direction.

8. Why this is not another scalar multiplier

It is tempting to replace the variational term by

$$h = 2\lambda K^\alpha g.$$

This is not the same mechanism. It preserves only the zeroth-order conformal part and hides the spatial derivative terms. Once K is inhomogeneous, the evolution of curvature includes Laplacians and Hessians of K^α. Those terms are of the same scale as the zeroth-order damping at a neck. The homogeneous success is therefore not stable.

The variational construction is different. The tensor

$$E_\Phi = \mathrm{grad}_{L^2} \int \Phi(|\mathrm{Rm}|^2) d\mu$$

contains the derivative terms forced by the variation of curvature. After DeTurck gauge fixing, this belongs to the same general class as higher-order curvature flows. The model therefore pays the price necessary for PDE honesty: fourth-order terms are present explicitly, not suppressed.

9. Relation to second-order uniqueness

One should not expect a universal second-order natural tensor to do this job. Second-order divergence-free natural symmetric tensors are severely restricted by Lovelock-type classification results. In low dimensions this leaves only metric and Einstein-type structures, not a new high-curvature damping channel. This supports, but does not by itself prove, the conclusion that a genuine local regularizer must leave the second-order class.

The present proposal leaves it by using a curvature-energy gradient.

10. What is proved algebraically

The construction gives an algebraically verified high-curvature regulator. It does not, by itself, give global regularity or noncollapse.

Theorem 1 (algebraic high-curvature verification)**.** *Let M^3 be compact and let $K = |\mathrm{Rm}|^2$. For the functional*

$$\mathcal{F}_q[g] = \int_M (K+\delta)^q \, d\mu_g$$

with $q > 3/4$, the negative $L^2(d\mu)$-gradient has the correct conformal high-curvature sign and produces a leading round-mode drift of order $-K^{q+1}$, which dominates the Ricci reaction $+K^{3/2}$. For the barrier functional

$$\mathcal{B}_{p,q}[g] = \int_M \frac{(K+\delta)^q}{(K_P-K)^p} \, d\mu_g$$

with $p > 0$, the same sign becomes singular as $K \nearrow K_P$.

Proof. The first assertion is the scale derivative computation with $\Phi(K) = (K+\delta)^q$. For $K \gg \delta$,

$$4K\Phi' - 3\Phi = (4q-3)K^q + o(K^q),$$

which is positive for $q > 3/4$. The induced conformal drift in K has an additional factor $-\varepsilon K/3$, giving order $-K^{q+1}$. Since $q+1 > 3/2$, this dominates $K^{3/2}$. The barrier assertion follows from

$$4K\Phi'(K) - 3\Phi(K) = \Phi(K)\left(\frac{4qK}{K+\delta} + \frac{4pK}{K_P - K} - 3\right),$$

whose second term diverges to $+\infty$ as $K \nearrow K_P$. □

11. Why noncollapse is a different problem

The curvature wall prevents a high-curvature channel. It cannot, by itself, prevent bounded-curvature collapse. The same Hopf–Berger calculation already shows the obstruction.

Lemma 2 (bounded-curvature collapse is invisible to K-walls)**.** *Let*

$$g_{1,c} = \sigma_1^2 + \sigma_2^2 + c^2\sigma_3^2$$

be the Berger family on S^3. As $c \downarrow 0$,

$$\mathrm{Vol}(g_{1,c}) \to 0, \qquad \mathrm{inj}(g_{1,c}) \to 0,$$

but

$$\sup_{S^3} |\mathrm{Rm}(g_{1,c})|^2$$

remains bounded. Hence no barrier depending only on the condition $K \nearrow K_P$ can be a complete noncollapse mechanism.

Proof. This is the $u = z/x \to 0$, $x = 1$ case of the curvature dictionary

$$|\mathrm{Rm}|^2 = \frac{4(16 - 24u + 11u^2)}{x^2}.$$

With $x = 1$, the right-hand side remains bounded and tends to 64. But the Hopf fibre length is proportional to c, hence the injectivity radius and volume collapse to zero. Thus bounded-curvature collapse is a separate channel. □

Remark 3 (geometric and spectral meaning of the Hopf collapse)**.** *The limit $c \downarrow 0$ is a Cheeger–Gromov collapse along the free Hopf S^1-action. The limiting metric tensor has vertical kernel and descends to the S^2 base of the Hopf fibration; it is not a high-curvature limit. This also explains why the first Laplace eigenvalue alone is not a reliable collapse detector. Fibre-invariant test functions have Rayleigh quotients controlled by the base geometry, while vertical modes become expensive as c^{-2}. Thus a useful spectral detector must be normalized or replaced by a heat-kernel, Sobolev, systolic, or injectivity-radius quantity.*

Proposition 3 (a heat-kernel detector sees fixed-base Hopf collapse)**.** *For $r > 0$, define*

$$\mathcal{H}_r(g) = \sup_{x \in M,\ 0 < t \le r^2} t^{3/2} H_g(t, x, x),$$

where H_g is the scalar heat kernel. Along the fixed-base Hopf collapse

$$g_{a,c} = a^2(\sigma_1^2 + \sigma_2^2) + c^2\sigma_3^2, \qquad a > 0, \qquad c \downarrow 0,$$

one has

$$\mathcal{H}_r(g_{a,c}) \to +\infty.$$

More precisely,

$$\mathcal{H}_r(g_{a,c}) \ge \frac{r^3}{\mathrm{Vol}(g_{a,c})} = \frac{r^3}{2\pi^2 a^2 c}.$$

Proof. For any compact Riemannian manifold, the zero eigenfunction gives the pointwise lower bound

$$H_g(t,x,x) \geq \frac{1}{\mathrm{Vol}(g)}$$

for every $t > 0$. Taking $t = r^2$ in the definition of $\mathcal{H}_r$ therefore gives

$$\mathcal{H}_r(g) \geq \frac{r^3}{\mathrm{Vol}(g)}.$$

For the Berger family used here,

$$\mathrm{Vol}(g_{a,c}) = 2\pi^2 a^2 c.$$

Thus

$$\mathcal{H}_r(g_{a,c}) \geq \frac{r^3}{2\pi^2 a^2 c} \to +\infty \qquad (c \downarrow 0).$$

□

Remark 4 (scope of the heat-kernel observation)**.** *Proposition 3 is deliberately modest. It proves that a heat-kernel scale detector sees the fixed-base Hopf collapse, whereas $\lambda_1(-\Delta)$ alone does not. It does not prove NC2 monotonicity for $\mathcal{H}_r$, and it does not replace the sharper adiabatic asymptotic*

$$H_{g_{a,c}}(t,x,x) \sim \frac{1}{4\pi t}\frac{1}{2\pi c} \qquad (c^2 \ll t \ll a^2),$$

whose rigorous proof requires either the exact Berger spectrum or a quoted adiabatic heat-kernel theorem. Thus the heat kernel is a natural non-additive collapse detector, not yet the missing noncollapse entropy.

Proposition 4 (Ricci lower bound from a wall gap)**.** *Let M^3 be a Riemannian three-manifold. If*

$$K = |\mathrm{Rm}|^2 \leq K_0,$$

then

$$\mathrm{Ric} \geq -\left(\frac{1}{2} + \frac{1}{\sqrt{3}}\right)\sqrt{K_0}\, g.$$

In particular, a wall gap

$$K \leq K_P - \eta$$

implies

$$\mathrm{Ric} \geq -\kappa_0 g, \qquad \kappa_0 = \left(\frac{1}{2} + \frac{1}{\sqrt{3}}\right)\sqrt{K_P - \eta}.$$

Proof. In dimension three,

$$K = 4|\mathrm{Ric}_0|^2 + \frac{1}{3}R^2.$$

Thus

$$|\mathrm{Ric}_0| \leq \frac{\sqrt{K}}{2}, \qquad |R| \leq \sqrt{3K}.$$

Every eigenvalue of the traceless symmetric tensor Ric_0 is bounded below by $-|\mathrm{Ric}_0|$, while

$$\mathrm{Ric} = \mathrm{Ric}_0 + \frac{R}{3}g.$$

Therefore

$$\lambda_{\min}(\mathrm{Ric}) \geq -|\mathrm{Ric}_0| - \frac{|R|}{3} \geq -\left(\frac{1}{2} + \frac{1}{\sqrt{3}}\right)\sqrt{K}.$$

The stated wall-gap estimate follows by inserting $K \leq K_P - \eta$. □

Corollary 1 (heat-kernel coercivity under a wall gap)**.** *Assume on a three-manifold that*

$$K \le K_P - \eta.$$

Then the constants in the local on-diagonal heat-kernel comparison estimates may be taken to depend only on K_P, η and the chosen scale. In particular, for $0 < \rho \le r$, standard Li–Yau/Davies type comparison under the Ricci lower bound of Proposition 4 gives a constant $c = c(K_P, \eta, r) > 0$ such that

$$H_g(\rho^2, x, x) \ge \frac{c}{\operatorname{Vol} B(x,\rho)}.$$

Consequently, if

$$\mathcal{H}_r(g) \le A,$$

then

$$\operatorname{Vol} B(x,\rho) \ge c\, A^{-1} \rho^3 \qquad (0 < \rho \le r).$$

Proof. By Proposition 4, the wall gap gives a uniform lower Ricci bound

$$\operatorname{Ric} \ge -\kappa_0 g, \qquad \kappa_0 = \left(\frac{1}{2} + \frac{1}{\sqrt{3}}\right)\sqrt{K_P - \eta}.$$

The local Li–Yau/Davies heat-kernel comparison theory under this lower Ricci bound gives the displayed on-diagonal lower estimate, with constants depending only on dimension, κ_0, and the scale r. On the other hand, $\mathcal{H}_r(g) \le A$ gives

$$H_g(\rho^2, x, x) \le A\rho^{-3}.$$

Combining the two inequalities yields

$$\frac{c}{\operatorname{Vol} B(x,\rho)} \le A\rho^{-3},$$

hence

$$\operatorname{Vol} B(x,\rho) \ge c\, A^{-1}\rho^3.$$

□

Remark 5 (meaning of the heat-kernel direction)**.** *Corollary 1 supplies an NC3-type coercivity direction for the heat-kernel scale detector, provided a wall gap is already known. It does not prove NC2 propagation of $\mathcal{H}_r$ along the curvature-wall flow. The remaining noncollapse problem is therefore the propagation estimate, or a replacement for it:*

$$\mathcal{H}_r(g(t)) \le C(T, K_P, \eta, r)\, \mathcal{H}_r(g(0)).$$

The known spectra of Berger spheres make this propagation question directly testable in the homogeneous sector, but such spectral probes are numerical evidence unless the normalization and the running supremum in $\mathcal{H}_r$ are treated as part of a separate argument.

Therefore a complete replacement for surgery cannot be only a local high-curvature wall. It must also contain a noncollapse principle.

12. NC1–NC3: the required noncollapse package

The noncollapse closure is encoded by three properties. These are stated here explicitly because they are not consequences of the local curvature-wall calculation.

Definition 2 (NC1–NC3)**.** *Let*

$$\Phi(K) = \frac{(K+\delta)^q}{(K_P - K)^p}, \qquad q > 3/4, \quad p > 0, \quad \delta > 0,$$

on

$$\mathcal{U}_P = \{g : \sup_M K < K_P\}.$$

A Perelman-type noncollapse functional for the barrier problem is a family $\mathcal{N}(g, f, \tau)$, with $f \in C^\infty(M)$, $\tau > 0$, satisfying:

NC1. Correct scale weight. *Under $g \mapsto \theta^2 g$, $\tau \mapsto \theta^2 \tau$, the barrier-coupling term has the unique power τ^{2q} in the pure model $\Phi(K) = K^q$.*

NC2. Monotonicity. *Along the relevant coupled flow of (g, f, τ), the minimized entropy is monotone in the Perelman direction.*

NC3. Coercivity implies noncollapse. *If the entropy is bounded below and $K \le r^{-4}$ on $B(x, r)$, with $r^{-4} < K_P$, then*

$$\operatorname{Vol} B(x, r) \ge \kappa(A, r)\, r^3.$$

The constant is allowed to depend on the fixed scale r, because K_P breaks exact scale invariance.

The first scale-correct candidate is

$$\mathcal{W}_\Phi(g, f, \tau) = \int_M \left[\tau(R + |\nabla f|^2) + \varepsilon \tau^{2q} \Phi(K) + f - 3\right] (4\pi\tau)^{-3/2} e^{-f}\, d\mu.$$

It is important to call this only a *candidate*. The power τ^{2q} is forced by scaling, but scaling does not imply monotonicity. For $\Phi = K^q$, NC1 is exact:

$$(\theta^2 \tau)^{2q} \Phi(\theta^{-4} K) = \theta^{4q} \tau^{2q} \theta^{-4q} K^q = \tau^{2q} K^q.$$

NC3 is soft once NC2 is known: the usual collapsed-ball test function gives $\mathcal{W}(g, f^*, \tau) \le \log \kappa_0 + C$, and the extra term is bounded on $K \le r^{-4} < K_P$. Thus $\mu_\Phi \to -\infty$ as $\kappa_0 \to 0$, contradicting an entropy lower bound. The entire hard content is NC2. The Berger calculation below shows that the natural eigenvalue version of this candidate does not satisfy NC2.

13. The NC2 obstruction for the unweighted barrier flow

Write

$$\mathcal{P}_\Phi[g] = \int_M \Phi(K)\, d\mu_g, \qquad K = |\mathrm{Rm}|^2,$$

and define E_Φ by

$$D\mathcal{P}_\Phi[g](h) = \int_M \langle E_\Phi, h\rangle\, d\mu.$$

With the curvature-variation conventions used here, the formal first-variation tensor has the Streets-type structure

$$(E_\Phi)_{ij} = \frac{1}{2}\Phi g_{ij} - 2\Phi'(K) R_i{}^{abc} R_{jabc} + 4\nabla^a \nabla^b \big(\Phi'(K) R_{iabj}\big).$$

For $\Phi(K) = K$ this is the usual L^2-curvature-gradient structure, up to the sign and index conventions of the chosen curvature tensor. In this paper the theorem-level consequences that use this tensor are checked on the homogeneous Berger family by symmetric criticality and by direct differentiation of the reduced functional. The full tensorial expression is recorded to explain the analytic structure of the flow; a complete PDE theorem would require an independent convention-by-convention comparison with the standard L^2-curvature-flow formula.

The bare barrier flow is

$$\partial_\tau g = -2\mathrm{Ric} - \varepsilon E_\Phi.$$

Perelman's entropy mechanism, however, is not naturally expressed in the unweighted inner product. It uses the weighted measure

$$\rho\, d\mu = (4\pi\tau)^{-3/2} e^{-f}\, d\mu.$$

Define the f-weighted Euler tensor E_Φ^f by

$$D \int_M \Phi(K) \rho\, d\mu = \int_M \langle E_\Phi^f, h\rangle\, \rho\, d\mu$$

with f and τ fixed during the metric variation. With the same convention, the corresponding weighted expression is

$$(E_\Phi^f)_{ij} = \frac{1}{2}\Phi g_{ij} - 2\Phi'(K) R_i{}^{abc} R_{jabc} + 4e^f \nabla^a \nabla^b \left(e^{-f} \Phi'(K) R_{iabj}\right).$$

Let

$$Q_{iabj} = \Phi'(K) R_{iabj}.$$

The discrepancy between the two expressions is the weighted-divergence correction

$$(E^f_\Phi - E_\Phi)_{ij} = 4\left[e^f \nabla^a \nabla^b (e^{-f} Q_{iabj}) - \nabla^a \nabla^b Q_{iabj}\right].$$

Expanding the weighted derivatives gives

$$\begin{aligned}(E^f_\Phi - E_\Phi)_{ij} = 4\big[&- (\nabla^a f)\nabla^b Q_{iabj} - (\nabla^b f)\nabla^a Q_{iabj} \\ &+ \big((\nabla^a f)(\nabla^b f) - \nabla^a \nabla^b f\big) Q_{iabj}\big].\end{aligned}$$

Lemma 3 (no pointwise sign for the discrepancy)**.** *The tensor $E^f_\Phi - E_\Phi$ has no intrinsic pointwise sign.*

Proof. At a point where $Q \neq 0$, choose normal coordinates and choose f so that $\nabla f = 0$ at that point while $\nabla^2 f$ is arbitrary. Then

$$(E^f_\Phi - E_\Phi)_{ij} = -4(\nabla^a \nabla^b f) Q_{iabj}.$$

Contracting with a symmetric tensor S^{ij} gives

$$S^{ij}(E^f_\Phi - E_\Phi)_{ij} = -4 S^{ij}(\nabla^a \nabla^b f) Q_{iabj}.$$

The Hessian of f may be chosen with either sign in any nonzero contraction direction. Thus the expression can be positive or negative. □

Consequently the original unweighted barrier flow has no automatic Perelman-type NC2 theorem. One must either prove a separate absorption estimate for this discrepancy or change the flow.

14. Berger falsification of the naive eigenvalue candidate

The preceding obstruction $E^f_\Phi - E_\Phi$ is not the only problem. The most natural eigenvalue candidate fails even in the homogeneous sector where that obstruction disappears.

Define

$$\lambda_\Phi(g) = \lambda_1(-4\Delta_g + R + \varepsilon\Phi(K)).$$

On a homogeneous Berger metric, R and K are spatially constant. Hence the ground state of $-4\Delta_g + R + \varepsilon\Phi(K)$ is the constant function and

$$\lambda_\Phi(g) = R + \varepsilon\Phi(K)$$

exactly. Thus f is constant, $\nabla f = 0$, and the weighted-gradient ambiguity is absent. This is the most favourable sector for a Perelman-type NC2 candidate.

Let

$$g_{a,c} = a^2(\sigma_1^2 + \sigma_2^2) + c^2\sigma_3^2, \qquad x = a^2, \quad z = c^2, \quad u = z/x.$$

The Berger curvature dictionary is

$$K = \frac{4(16x^2 - 24xz + 11z^2)}{x^4}, \qquad R = \frac{8}{x} - \frac{2z}{x^2},$$

and the unforced Ricci-flow ODE is

$$\dot{x}_0 = -8 + \frac{4z}{x}, \qquad \dot{z}_0 = -\frac{4z^2}{x^2}.$$

At $u = 1$, $R = 6/x$, $K = 12/x^2$, and

$$C_3^{\mathrm{B}} = \frac{\dot{K}}{K^{3/2}} = \frac{4}{\sqrt{3}}.$$

Proposition 5 (explicit Berger form of E_Φ)**.** *For any smooth Φ, on the Berger family*

$$g_{a,c} = a^2(\sigma_1^2 + \sigma_2^2) + c^2\sigma_3^2, \qquad x = a^2, \quad z = c^2,$$

with

$$K = \frac{4(16x^2 - 24xz + 11z^2)}{x^4},$$

the symmetric-criticality reduction of $E_\Phi = \operatorname{grad}_{L^2} \int \Phi(K) d\mu$ is

$$\boxed{(E_\Phi)_{11} = -\frac{8\Phi'(K)(8x^2 - 18xz + 11z^2)}{x^3} + \frac{x}{2}\Phi(K),}$$

and

$$\boxed{(E_\Phi)_{33} = -\frac{8z^2\Phi'(K)(12x - 11z)}{x^4} + \frac{z}{2}\Phi(K).}$$

At the round point $z = x$, both components reduce to

$$(E_\Phi)_{11} = (E_\Phi)_{33} = -\frac{8\Phi'(K)}{x} + \frac{x}{2}\Phi(K),$$

as required by the enhanced isotropy of the round metric.

Proof. The reduced functional is

$$\mathcal{P}_\Phi(x, z) = \Phi(K(x, z)) \operatorname{Vol}(x, z).$$

By symmetric criticality,

$$(E_\Phi)_{11} = \frac{x^2}{2\operatorname{Vol}}\partial_x \mathcal{P}_\Phi, \qquad (E_\Phi)_{33} = \frac{z^2}{\operatorname{Vol}}\partial_z \mathcal{P}_\Phi.$$

Since

$$K_x = -\frac{16(8x^2 - 18xz + 11z^2)}{x^5}, \qquad K_z = -\frac{8(12x - 11z)}{x^4},$$

and

$$\frac{\partial_x \operatorname{Vol}}{\operatorname{Vol}} = \frac{1}{x}, \qquad \frac{\partial_z \operatorname{Vol}}{\operatorname{Vol}} = \frac{1}{2z},$$

the displayed formulas follow immediately. For $\Phi(K) = K$ they reduce to the components used below. □

Corollary 2 (Planck-wall components and their signs)**.** *For the physical wall*

$$\Phi(K) = \frac{(K + \delta)^q}{(K_P - K)^p}, \qquad p > 0, \quad q > 0, \quad \delta > 0,$$

put

$$A(K) = \frac{\Phi'(K)}{\Phi(K)} = \frac{q}{K + \delta} + \frac{p}{K_P - K}.$$

Then Proposition 5 becomes

$$\boxed{(E_\Phi)_{11} = \Phi(K)\left[\frac{x}{2} - \frac{8}{x}(11u^2 - 18u + 8)A(K)\right],}$$

and

$$\boxed{(E_\Phi)_{33} = \Phi(K)\left[\frac{ux}{2} - \frac{8u^2}{x}(12 - 11u)A(K)\right].}$$

Since

$$11u^2 - 18u + 8 > 0$$

for all real u, the horizontal component is negative whenever

$$x^2 < 16(11u^2 - 18u + 8)A(K).$$

In particular, as $K \nearrow K_P$, one has $A(K) \to +\infty$, and therefore

$$(E_\Phi)_{11} < 0$$

near the wall.

For the vertical component, if $u > 12/11$, then $12 - 11u < 0$, and hence

$$(E_\Phi)_{33} > 0.$$

If $0 < u < 12/11$, then

$$(E_\Phi)_{33} < 0 \quad \Longleftrightarrow \quad x^2 < 16u(12 - 11u)A(K).$$

Consequently, near the wall,

$$(E_\Phi)_{33} < 0 \quad (0 < u < 12/11), \qquad (E_\Phi)_{33} > 0 \quad (u > 12/11).$$

Thus the wall has a definite leading sign structure on the Berger sectors: it expands horizontal directions near the wall, while the vertical reaction changes sign at $u = 12/11$.

Proposition 6 (reduced Berger dynamics for a general wall)**.** *On the Berger family, the curvature-wall equation*

$$\dot{g} = -2\mathrm{Ric} - \varepsilon E_\Phi$$

reduces to

$$\boxed{\dot{x} = -8 + 4u + \frac{8\varepsilon\Phi'(K)}{x}(8 - 18u + 11u^2) - \frac{\varepsilon x}{2}\Phi(K),}$$

and

$$\boxed{\dot{z} = -4u^2 + \frac{8\varepsilon u^2\Phi'(K)}{x}(12 - 11u) - \frac{\varepsilon u x}{2}\Phi(K).}$$

Consequently, for $u = z/x$,

$$\boxed{\dot{u} = \frac{8u(1-u)}{x}\left(1 + \frac{2\varepsilon\Phi'(K)}{x}(11u - 4)\right).}$$

In the anisotropy equation the zeroth-order $\Phi(K)$-terms cancel; only the slope $\Phi'(K)$ of the wall affects the fibre ratio.

Proof. The x, z-system is obtained by substituting Proposition 5 into

$$\dot{x} = \dot{x}_0 - \varepsilon(E_\Phi)_{11}, \qquad \dot{z} = \dot{z}_0 - \varepsilon(E_\Phi)_{33}.$$

Since $z = ux$, one has

$$\dot{u} = \frac{\dot{z} - u\dot{x}}{x}.$$

Substitution gives the displayed formula. The cancellation of the $\Phi(K)$-terms follows from their pure scaling character:

$$-\frac{\varepsilon u x}{2}\Phi(K) - u\left(-\frac{\varepsilon x}{2}\Phi(K)\right) = 0.$$

□

Corollary 3 (Berger cone and Hopf-sector warning)**.** *Assume $\varepsilon > 0$ and $\Phi'(K) > 0$. Then, on the Berger sector,*

$$\frac{4}{11} \le u \le M, \qquad M > 1,$$

is forward invariant. More precisely, $u = 1$ is invariant; for $\frac{4}{11} \le u < 1$ one has $\dot{u} > 0$, and for $u > 1$ one has $\dot{u} < 0$. In the lower Hopf sector $0 < u < 4/11$, the wall can instead push u downward whenever

$$\boxed{\frac{2\varepsilon\Phi'(K)}{x}(4 - 11u) > 1.}$$

Thus a K-wall controls high curvature but need not oppose every bounded-curvature fibre-collapse direction.

Proof. The sign of $\dot{u}$ is the sign of

$$u(1-u)\left(1+\frac{2\varepsilon\Phi'(K)}{x}(11u-4)\right).$$

If $u \geq 4/11$, the parenthesis is at least 1. The stated signs follow. If $0 < u < 4/11$, the parenthesis is negative precisely under the displayed inequality. □

Corollary 4 (curvature, scale, and volume consequences)**.** *Along the Berger family,*

$$K = \frac{4(16-24u+11u^2)}{x^2}.$$

Hence a lower bound $x \geq x_0 > 0$ gives a curvature bound only together with an upper bound for u. In particular an a-barrier $a^2 = x \geq x_0$ is not equivalent to the invariant curvature wall $K < K_P$, since $u \to \infty$ at fixed x makes $K \to \infty$.

The volume satisfies

$$\mathrm{Vol}(g_{a,c}) = 2\pi^2 x\sqrt{z}$$

and

$$\boxed{\frac{d}{dt}\log \mathrm{Vol} = \frac{-8+2u}{x} + \frac{4\varepsilon\Phi'(K)}{x^2}(16-24u+11u^2) - \frac{3\varepsilon}{4}\Phi(K).}$$

On the round locus $u = 1$, this becomes

$$\boxed{\frac{d}{dt}\log \mathrm{Vol} = -\frac{6}{x} + \frac{12\varepsilon\Phi'(K)}{x^2} - \frac{3\varepsilon}{4}\Phi(K).}$$

Volume normalization therefore does not by itself remove Hopf-fibre collapse: at fixed volume $x\sqrt{z} = \mathrm{const}$, the limit $z \downarrow 0$ is still a fibre collapse, with $x \to \infty$.

Proof. The curvature formula is the Berger dictionary. The statement about x and u is immediate from the displayed expression for K. For the volume,

$$\frac{d}{dt}\log \mathrm{Vol} = \frac{\dot{x}}{x} + \frac{1}{2}\frac{\dot{z}}{z}.$$

Substituting Proposition 6 gives the displayed formula. □

Corollary 5 (normalization, scalar curvature, and traceless anisotropy)**.** *Volume normalization changes only the common scale in the Berger reduction. Indeed, replacing a velocity v by*

$$v^{\mathrm{norm}} = v - \frac{1}{3}\overline{\mathrm{tr}_g v}\, g$$

changes

$$\dot{x} \mapsto \dot{x} - \rho x, \qquad \dot{z} \mapsto \dot{z} - \rho z$$

for a scalar $\rho(t)$, and therefore leaves

$$u = \frac{z}{x}$$

unchanged:

$$\boxed{\dot{u}_{\mathrm{norm}} = \dot{u}_{\mathrm{unnorm}}.}$$

Thus volume normalization cannot by itself remove Hopf-fibre collapse; it changes the common scale but not the anisotropy equation.

The scalar curvature and traceless Ricci size are

$$R = \frac{8-2u}{x}, \qquad \boxed{|\mathrm{Ric}_0|^2 = \frac{32(1-u)^2}{3x^2}.}$$

Along the general Berger wall-flow,

$$\boxed{\dot{R} = \frac{\varepsilon\Phi(K)x^2(4-u) + 16\varepsilon\Phi'(K)(3u-4)(11u^2-16u+8) + 8x(3u^2-8u+8)}{x^3}.}$$

At $u = 1$, this reduces to

$$\boxed{\dot{R} = \frac{24}{x^2} + \frac{3\varepsilon\Phi(K)}{x} - \frac{48\varepsilon\Phi'(K)}{x^3}.}$$

The dimensionless shape anisotropy decreases in the protected cone: if $u \geq 4/11$, $u \neq 1$, $\varepsilon > 0$, and $\Phi'(K) > 0$, then

$$\boxed{\frac{d}{dt}(1-u)^2 < 0.}$$

The absolute quantity $|\mathrm{Ric}_0|^2$, however, can still grow if the scale x decreases fast enough. Thus the Berger dynamics separates shape rounding from absolute curvature growth.

Proof. The normalization statement follows because subtracting a scalar multiple of the metric subtracts the same logarithmic rate from x and z. The formula for R is the Berger curvature dictionary, and the identity

$$K = 4|\mathrm{Ric}_0|^2 + \frac{1}{3}R^2$$

gives the displayed expression for $|\mathrm{Ric}_0|^2$. Differentiating $R = (8-2u)/x$ using Proposition 6 gives the formula for $\dot{R}$. Finally,

$$\frac{d}{dt}(1-u)^2 = -\frac{16u(1-u)^2}{x}\left(1 + \frac{2\varepsilon\Phi'(K)}{x}(11u-4)\right),$$

whose sign is negative in the stated cone. □

Proposition 7 (exact Berger falsification for $\Phi(K) = K$)**.** *For the pure quartic coupling $\Phi(K) = K$, the reduced Euler tensor gives*

$$(E_\Phi)_{11} = \frac{2(-16x^2 + 48xz - 33z^2)}{x^3}, \qquad (E_\Phi)_{33} = \frac{2z(16x^2 - 72xz + 55z^2)}{x^4}.$$

Along the forced Berger system

$$\dot{x} = \dot{x}_0 - \varepsilon(E_\Phi)_{11}, \qquad \dot{z} = \dot{z}_0 - \varepsilon(E_\Phi)_{33},$$

one has

$$\boxed{\dot{\lambda}_\Phi(x,u,\varepsilon) = \frac{8c_0(u)}{x^2} - \frac{4c_1(u)}{x^3}\varepsilon - \frac{16c_2(u)}{x^4}\varepsilon^2,}$$

where

$$c_0(u) = 3u^2 - 8u + 8,$$
$$c_1(u) = 143u^3 - 436u^2 + 464u - 192,$$
$$c_2(u) = 1331u^4 - 3696u^3 + 3648u^2 - 1536u + 256.$$

At the round point $u = 1$,

$$\boxed{\dot{\lambda}_\Phi\big|_{u=1} = \frac{12(2x-\varepsilon)(x+4\varepsilon)}{x^4}.}$$

Proof. The reduced functional is

$$\mathcal{P}_\Phi(x,z) = \Phi(K(x,z))\ \mathrm{Vol}(x,z).$$

By symmetric criticality,

$$(E_\Phi)_{11} = \frac{x^2}{2\,\mathrm{Vol}}\,\partial_x\mathcal{P}_\Phi, \qquad (E_\Phi)_{33} = \frac{z^2}{\mathrm{Vol}}\,\partial_z\mathcal{P}_\Phi.$$

For $\Phi = K$, direct differentiation gives the displayed components of E_Φ. Substituting them into the forced ODE and differentiating

$$\lambda_\Phi = R + \varepsilon K$$

gives the stated formula. The round specialization $z = x$ simplifies to

$$\dot{\lambda}_\Phi = \frac{24x^2 + 84x\varepsilon - 48\varepsilon^2}{x^4} = \frac{12(2x-\varepsilon)(x+4\varepsilon)}{x^4}.$$

□

Corollary 6 (the naive NC2 candidate is false)**.** *For every fixed $\varepsilon > 0$,*

$$\dot{\lambda}_\Phi\big|_{u=1} < 0 \qquad \text{whenever} \qquad x < \frac{\varepsilon}{2}.$$

Thus $\lambda_\Phi = R + \varepsilon K$ is not monotone along the barrier flow. This is not a failure of an estimate; it is an exact sign reversal.

Remark 6 (the sign reversal is not only round)**.** *The polynomial $c_0(u) = 3u^2 - 8u + 8$ is positive for all $u > 0$, since its discriminant is negative. The polynomial $c_2(u)$ has no positive root by Sturm's test and $c_2(0) = 256 > 0$. Hence the negative $-16c_2(u)\varepsilon^2/x^4$ term eventually dominates for every fixed fibre ratio $u > 0$. The round calculation is only the simplest visible instance.*

Proposition 8 (the Planck wall worsens the sign)**.** *Let*

$$\Phi(K) = \frac{(K+\delta)^q}{(K_P - K)^p}, \qquad p > 0, \quad q > 0, \quad \delta > 0.$$

At the round point, along the same unweighted barrier flow,

$$\dot{\lambda}_\Phi \longrightarrow -\infty \qquad \text{as} \qquad K \nearrow K_P$$

for every fixed $\varepsilon > 0$.

Proof. At the round point,

$$R = \frac{6}{x}, \qquad K = \frac{12}{x^2}.$$

For a pure scaling variation,

$$\dot{x} = -4 + \frac{\varepsilon x}{6}\big(4K\Phi'(K) - 3\Phi(K)\big).$$

Since

$$\dot{R} = -\frac{R}{x}\dot{x}, \qquad \dot{K} = -\frac{2K}{x}\dot{x},$$

we get

$$\begin{aligned}\dot{\lambda}_\Phi &= \dot{R} + \varepsilon\Phi'(K)\dot{K} \\ &= \frac{4R}{x} + \frac{8\varepsilon K\Phi'(K)}{x} - \varepsilon\left(\frac{R}{6} + \frac{\varepsilon K\Phi'(K)}{3}\right)\big(4K\Phi'(K) - 3\Phi(K)\big).\end{aligned}$$

Write $s = K_P - K$. As $s \downarrow 0$,

$$\Phi(K) \sim Cs^{-p}, \qquad \Phi'(K) \sim pCs^{-p-1},$$

with $C = (K_P + \delta)^q > 0$. Therefore

$$4K\Phi'(K) - 3\Phi(K) \sim 4pKC\, s^{-p-1},$$

and the leading term in $\dot{\lambda}_\Phi$ is

$$-\frac{\varepsilon^2 K\Phi'(K)}{3}\big(4K\Phi'(K) - 3\Phi(K)\big) \sim -\frac{4\varepsilon^2 p^2 K^2 C^2}{3}\, s^{-2p-2}.$$

All positive terms have lower order s^{-p-1}. Hence $\dot{\lambda}_\Phi \to -\infty$. □

The conclusion is sharp:

$$\boxed{\lambda_\Phi = R + \varepsilon\Phi(K) \text{ is not an NC2 functional.}}$$

The curvature wall still functions as a local metric damping term, but the eigenvalue/entropy route $R + \varepsilon\Phi(K)$ is closed. Any future noncollapse functional must be structurally different and must first pass this Berger test.

15. No-go for additive matching K-potential repairs

The preceding Berger falsification closes the most direct eigenvalue route. A broader additive class also fails. The following result is deliberately stated only for the class that is actually tested: an additive K-potential is inserted into the Perelman functional, and the reduced metric forcing is generated by the same added scalar through symmetric criticality. It does not claim a no-go theorem for arbitrary smooth functions $H(\tau, K)$.

On the round Berger locus $g_x = x g_{\text{round}}$,

$$R = \frac{6}{x}, \qquad K = \frac{12}{x^2}.$$

Let

$$\mathcal{N}_F = \mathcal{W}_{\text{Perelman}} + \sigma\varepsilon\tau^\alpha K, \qquad \sigma \in \{\pm 1\}, \quad \alpha \in \mathbb{R}.$$

At the Berger level, the Ricci part is still the Ricci-flow part; only the added K-term contributes its matching gradient forcing. This avoids confusing the tensor inside Perelman's monotonicity square,

$$\mathrm{Ric} + \nabla^2 f - \frac{1}{2\tau} g,$$

with the actual Ricci-flow velocity $-2\mathrm{Ric}$.

Theorem 2 (Berger no-go for additive matching K-potentials)**.** *For the pure representative $\Phi(K) = K$, at the round point and with $\tau = 1$,*

$$\frac{d}{dt}\mathcal{N}_F = \frac{3N(x)}{2x^4},$$

where

$$N(x) = x^2(x-4)^2 + \sigma\varepsilon(2 - 8\alpha)x^2 + 56\sigma\varepsilon x - 32\varepsilon^2.$$

Consequently, for every $\varepsilon > 0$, every sign σ, and every exponent α, the derivative of $\mathcal{N}_F$ is negative for some smooth round Berger initial data. In particular,

$$N(0) = -32\varepsilon^2 < 0.$$

Thus no member of this additive matching K-potential class satisfies NC2.

Proof. The first term $x^2(x-4)^2$ is the classical Perelman square in the round reduction. The remaining terms are obtained by differentiating the added scalar $\sigma\varepsilon\tau^\alpha K$ along the reduced matching-gradient forcing and then setting $z = x$, $\tau = 1$.

The displayed formula gives

$$N(0) = -32\varepsilon^2,$$

which is independent of σ and α. By continuity there exists $x_0 > 0$ such that $N(x) < 0$ for $0 < x < x_0$. Choosing a smooth round initial metric with such an x gives

$$\left.\frac{d}{dt}\mathcal{N}_F\right|_{t=0} < 0.$$

Monotonicity of an NC2 functional must hold for all smooth initial data in the tested class. Hence the additive matching K-potential class fails the Berger test. □

Theorem 3 (analytic near-wall no-go for Planck-barrier additive repairs)**.** *Let*

$$\Phi(K) = \frac{(K+\delta)^q}{(K_P - K)^p}, \qquad p > 0, \quad q > 0, \quad \delta > 0.$$

Consider the additive matching-potential class

$$\mathcal{N}_{\Phi,\alpha} = \mathcal{W}_{\text{Perelman}} + \sigma\varepsilon\tau^\alpha\Phi(K), \qquad \sigma \in \{\pm 1\}, \quad \alpha \in \mathbb{R},$$

with the reduced metric forcing generated by the same added scalar. At the round Berger point, with $\tau = 1$, write

$$s = K_P - K, \qquad x_P = \sqrt{\frac{12}{K_P}}.$$

Then, as $s \downarrow 0$,

$$\boxed{\frac{d}{dt}\mathcal{N}_{\Phi,\alpha} = -\frac{192\,p^2(K_P+\delta)^{2q}}{x_P^4}\,\varepsilon^2\,s^{-2p-2} + O_{\varepsilon,\sigma,\alpha}(s^{-p-1}).}$$

In particular,

$$\frac{d}{dt}\mathcal{N}_{\Phi,\alpha} \longrightarrow -\infty \qquad as \qquad K \nearrow K_P$$

for every fixed $\varepsilon \neq 0$, independent of the sign σ and of the exponent α.

Proof. On the round Berger locus, the classical part of Perelman's functional gives the usual square. Differentiating the additive potential along the matching-gradient reduced forcing and organizing the result by powers of ε gives

$$\frac{d}{dt}\mathcal{N}_{\Phi,\alpha} = \frac{3(x-4)^2}{2x^2} + \sigma\varepsilon A_{\Phi,\alpha}(x) + \varepsilon^2\frac{12\Phi'(K)\big(\Phi(K)x^2 - 16\Phi'(K)\big)}{x^4},$$

where $A_{\Phi,\alpha}(x) = O(s^{-p-1})$ as $s = K_P - K \downarrow 0$. The exact form of $A_{\Phi,\alpha}$ depends on the sign and τ-weight, but it is linear in the coupling and lower order at the wall.

Now

$$\Phi(K) \sim (K_P+\delta)^q s^{-p},$$

and

$$\Phi'(K) = q(K+\delta)^{q-1}s^{-p} + p(K+\delta)^q s^{-p-1} \sim p(K_P+\delta)^q s^{-p-1}.$$

Hence

$$\Phi(K)\Phi'(K) = O(s^{-2p-1}), \qquad \Phi'(K)^2 \sim p^2(K_P+\delta)^{2q}s^{-2p-2}.$$

Therefore the quadratic term has leading part

$$\varepsilon^2\frac{12\Phi'(K)(\Phi(K)x^2 - 16\Phi'(K))}{x^4} \sim -\frac{192p^2(K_P+\delta)^{2q}}{x_P^4}\varepsilon^2 s^{-2p-2}.$$

This dominates both the classical square and the linear coupling term. The leading coefficient is strictly negative and contains ε^2, so it is independent of the sign of the additive coupling. The theorem follows. □

Remark 7 (scope of the no-go theorems)**.** *The preceding two results close the matching additive class*

$$\mathcal{W}_{\mathrm{Perelman}} + \sigma\varepsilon\tau^\alpha K$$

in the bulk round-collapse limit, and the corresponding matching Planck-wall additive class at $K = K_P$. They do not close arbitrary smooth additive terms $H(\tau, K)$; for such a term independent H_τ and H_K contributions enter the Berger derivative and require a separate admissibility hypothesis or a separate no-go theorem. Nor do they rule out genuinely non-additive noncollapse mechanisms.

16. The unmodified Perelman entropy passes the protected Berger cone

The no-go theorems above concern additive repairs of Perelman's entropy by a matching K-potential. They do not test the unmodified Perelman entropy itself as an independent noncollapse detector. On the Berger sector the unmodified entropy passes the natural two-mechanism test.

Let

$$\nu_{\mathrm{B}}(g) = \inf_{\tau>0} \mathcal{W}_{\mathrm{Perelman}}(g, f, \tau)$$

with f restricted to the invariant density. On a homogeneous Berger metric this density is constant. In the positive-scalar region

$$R = \frac{8-2u}{x} > 0 \quad \Longleftrightarrow \quad u < 4,$$

the minimising scale is

$$\tau = \frac{3}{2R}.$$

Using

$$\mathrm{Vol}(g_{a,c}) = 2\pi^2 x^{3/2}\sqrt{u}, \qquad R = \frac{8-2u}{x},$$

one obtains, up to an additive universal constant,

$$\nu_{\mathrm{B}}(x,u) = \log \mathrm{Vol}(g_{a,c}) + \frac{3}{2}\log R + C = \frac{1}{2}\log u + \frac{3}{2}\log(8-2u) + C.$$

Theorem 4 (Berger two-mechanism test)**.** *Assume $\varepsilon > 0$ and $\Phi'(K) > 0$. On the protected positive-scalar Berger cone*

$$\frac{4}{11} \le u < 4,$$

the unmodified Perelman entropy reduction ν_{B} is monotone along the curvature-wall Berger flow. More precisely,

$$\frac{d}{dt}\nu_{\mathrm{B}} = \frac{32(1-u)^2}{x(8-2u)}\left(1 + \frac{2\varepsilon\Phi'(K)}{x}(11u-4)\right) \ge 0,$$

with equality only at the round point $u = 1$.

Proof. Differentiating the reduced entropy gives

$$\frac{\partial \nu_{\mathrm{B}}}{\partial u} = \frac{1}{2u} - \frac{3}{8-2u} = \frac{4(1-u)}{u(8-2u)}.$$

The exact Berger anisotropy equation for the wall flow is

$$\dot{u} = \frac{8u(1-u)}{x}\left(1 + \frac{2\varepsilon\Phi'(K)}{x}(11u-4)\right).$$

Multiplying these two expressions gives the stated derivative formula. In the cone $u \ge 4/11$, the bracket is at least one. Since $u < 4$, the denominator $8-2u$ is positive. Hence $d\nu_{\mathrm{B}}/dt \ge 0$, and equality occurs only when $u = 1$.

For the pure quartic wall $\Phi(K) = K$, this reduces to the fully factored form

$$\frac{d}{dt}\nu_{\mathrm{B}} = \frac{16(u-1)^2\big[x + 2\varepsilon(11u-4)\big]}{x^2(4-u)}.$$

For the physical Planck wall

$$\Phi(K) = \frac{(K+\delta)^q}{(K_P - K)^p},$$

one has $\Phi'(K) > 0$ on $K < K_P$. Therefore the same monotonicity conclusion holds throughout the protected positive-scalar cone. □

Corollary 7 (Berger noncollapse implication under a wall gap)**.** *On the protected positive-scalar Berger cone, suppose in addition that a wall gap holds:*

$$K \le K_P - \eta.$$

Then

$$|R| \le \sqrt{3(K_P - \eta)}.$$

Consequently, on every scale

$$0 < r \le 3^{-1/4}(K_P - \eta)^{-1/4},$$

one has $R \le r^{-2}$. Since $\nu_{\mathrm{B}}(g(t)) \ge \nu_{\mathrm{B}}(g(0))$, Perelman's static noncollapse implication gives

$$\mathrm{Vol}\, B(x,r) \ge \kappa(\nu_{\mathrm{B}}(g(0)))\, r^3$$

on those scales.

Remark 8 (scope)**.** *This is the first positive non-additive Berger result in the present programme. It says that the curvature wall and the unmodified Perelman entropy pass the homogeneous two-mechanism test:*

curvature wall controls high curvature, ν_{B} *controls collapse in the Berger reduction.*

It is not yet a full PDE monotonicity theorem for Perelman's ν *along the inhomogeneous DeTurck-gauged wall flow. The missing analytic statement is a propagation estimate, or monotonicity formula, for the unmodified Perelman entropy under the actual fourth-order forced flow.*

Remark 9 (no PDE envelope identity is used here)**.** *The Berger theorem above is not derived from a formal full-PDE identity for* $D\nu[g]$*. It uses only the explicit homogeneous minimisation over the constant density and over* τ*, followed by direct differentiation of the reduced invariant along the verified Berger ODE.*

This distinction is important. A naive formula of the form

$$D\nu_g[h] = -\int_M \left\langle \mathrm{Ric} + \nabla^2 f - \frac{1}{2\tau} g,\ h \right\rangle d\nu$$

is not used in this paper. If f *and* τ *are frozen during the metric variation, trace and scale terms appear, and a full envelope-theorem derivation must account for the Euler–Lagrange equations in* f *and* τ *with the chosen normalization. The inhomogeneous PDE monotonicity problem for the unmodified entropy is therefore left as a separate analytic problem.*

17. A PDE programme for the unmodified Perelman entropy

The Berger theorem proves that the unmodified Perelman entropy is monotone on the protected positive-scalar Berger cone. The full inhomogeneous PDE statement remains open. This section records the precise analytic and algebraic problems that must be solved in order to pass from the homogeneous theorem to a genuine PDE monotonicity or lower-bound propagation theorem.

Let M^3 be closed, and let $g(t)$ solve the DeTurck-gauged wall flow

$$\partial_t g = -2\mathrm{Ric} - \varepsilon E_\Phi + \mathcal{L}_W g, \tag{1}$$

where

$$E_\Phi = \mathrm{Grad}_{L^2} \int_M \Phi(|\mathrm{Rm}|^2)\, d\mu, \qquad \Phi(K) = \frac{(K+\delta)^q}{(K_P - K)^p}.$$

Assume on the time interval under consideration that the wall gap holds,

$$K = |\mathrm{Rm}|^2 \le K_P - \eta,$$

and that the scalar curvature is positive. Let (f, τ) be a minimising pair for Perelman's ν-functional at $g(t)$, and write

$$d\nu = (4\pi\tau)^{-3/2} e^{-f}\, d\mu, \qquad T_{ij} = \mathrm{Ric}_{ij} + \nabla_i \nabla_j f - \frac{1}{2\tau} g_{ij}.$$

At a differentiability point of ν, and for a smooth minimising branch (f, τ), the exact first variation is

$$D\nu_g[h] = -\tau \int_M \langle h, T \rangle\, d\nu.$$

The DeTurck term contributes nothing, since ν is diffeomorphism-invariant. Therefore the geometric part of the wall flow gives the exact identity

$$\boxed{\frac{d}{dt}\nu(g(t)) = 2\tau \int_M |T|^2\, d\nu + \varepsilon\tau \int_M \langle E_\Phi, T \rangle\, d\nu.} \tag{2}$$

Thus the whole PDE monotonicity problem is concentrated in the wall cross term

$$\mathfrak{C}_\Phi(g, f, \tau) = \int_M \langle E_\Phi, T \rangle\, d\nu.$$

The minimiser identities give the weighted divergence relation

$$\operatorname{div}_f T = 0, \qquad (\operatorname{div}_f T)_i = \nabla^j T_{ij} - T_{ij}\nabla^j f,$$

and the averaged trace identity

$$\int_M \operatorname{tr} T \, d\nu = 0.$$

These are weighted identities. They do not imply that T is pointwise trace-free, and they do not imply ordinary divergence-freeness.

The first open task is to compute $\mathfrak{C}_\Phi$ exactly from the unweighted wall tensor

$$(E_\Phi)_{ij} = \frac{1}{2}\Phi g_{ij} - 2\Phi'(K) R_i{}^{abc} R_{jabc} + 4\nabla^a\nabla^b\big(\Phi'(K) R_{iabj}\big),$$

with the convention caveat already stated in the paper. Since the integral is weighted by $d\nu$, the double-divergence term must be integrated by parts using the weighted adjoint. This produces terms involving

$$\nabla f, \qquad \nabla^2 f, \qquad \nabla\Phi'(K), \qquad \nabla^2\Phi'(K),$$

and curvature commutators. Under the wall gap, the singular coefficients Φ', Φ'', and Φ''' are bounded by constants depending on η, but the corresponding derivative terms still have to be estimated.

The second open task is algebraic. One must find a tensorial pinching condition replacing the Berger condition

$$u \geq \frac{4}{11}.$$

A naive condition such as

$$|\mathrm{Ric}_0| \leq cR, \qquad R > 0,$$

does not automatically reproduce the full protected Berger cone. The constant c must be checked against the Berger model, and the resulting condition must then be tested for preservation under the fourth-order wall flow.

The desired estimate is either the monotonicity estimate

$$\mathfrak{C}_\Phi(g, f, \tau) \geq 0,$$

or the weaker absorbable bound

$$\mathfrak{C}_\Phi(g, f, \tau) \geq -\theta \int_M |T|^2 \, d\nu, \qquad \varepsilon\theta < 2.$$

The first estimate would give

$$\frac{d}{dt}\nu(g(t)) \geq 0.$$

The second would still give monotonicity after absorption:

$$\frac{d}{dt}\nu(g(t)) \geq (2 - \varepsilon\theta)\tau \int_M |T|^2 \, d\nu \geq 0.$$

A still weaker but useful target is an integrated negative-part estimate,

$$\int_0^T \big(\mathfrak{C}_\Phi(g(t), f_t, \tau_t)\big)_- \, dt \leq C(T),$$

which would imply

$$\nu(g(t)) \geq \nu(g(0)) - C(T).$$

This lower-bound propagation is sufficient for finite-time noncollapse, even without strict monotonicity.

The zeroth-order contribution

$$\frac{1}{2}\int_M \Phi(K) \operatorname{tr} T \, d\nu$$

does not vanish automatically, since only $\int_M \operatorname{tr} T \, d\nu = 0$ is known. The factor $\Phi(K)$ prevents immediate cancellation. This term must be kept in the cross-term calculation.

Likewise, the curvature term

$$-2\int_M \Phi'(K)\, T^{ij} R_i{}^{abc} R_{jabc}\, d\nu$$

has no sign *a priori*. Its sign or absorbability must be proved from the same pinching condition that replaces the Berger inequality.

Finally, any PDE monotonicity theorem must be coupled to the analytic theory of the wall flow: DeTurck parabolicity, short-time existence, preservation of the wall gap, regularity estimates, and preservation of the required pinching condition.

Once monotonicity or lower-bound propagation of ν is established, noncollapse follows by Perelman's static theorem. Indeed, the wall gap implies

$$|R| \le \sqrt{3(K_P - \eta)}.$$

Thus the scalar-curvature hypothesis in the static noncollapse theorem is satisfied on sufficiently small scales depending only on K_P and η. A uniform lower bound for $\nu(g(t))$ then yields

$$\operatorname{Vol} B(x,r) \ge \kappa\, r^3$$

on those scales.

The status is therefore:

$$\boxed{\text{Berger two-mechanism theorem: proved;}}$$

$$\boxed{\text{full PDE monotonicity of unmodified } \nu\text{: open;}}$$

$$\boxed{\text{finite-time noncollapse follows from any propagated lower bound for } \nu.}$$

Any solution of the full PDE problem must first reproduce the Berger factorisation in the homogeneous limit. The Berger theorem is therefore not merely evidence; it is the consistency test for every proposed inhomogeneous identity.

18. Analytic roadmap for the DeTurck-gauged Planck-wall flow

We record the analytic structure of the curvature-wall flow. This section is not a full PDE continuation theorem. It explains what a future PDE closure would have to prove.

The ungauged flow is

$$\partial_t g = -2\mathrm{Ric} - \varepsilon E_\Phi, \qquad E_\Phi = \mathrm{Grad}_{L^2} \int_M \Phi(K)\, d\mu, \qquad K = |\mathrm{Rm}|^2,$$

where

$$\Phi(K) = \frac{(K+\delta)^q}{(K_P - K)^p}, \qquad p > 0, \quad q > 0, \quad \delta > 0.$$

With the conventions of this paper, the formal first-variation tensor has the Streets-type structure

$$(E_\Phi)_{ij} = \frac{1}{2}\Phi g_{ij} - 2\Phi'(K) R_i{}^{abc} R_{jabc} + 4\nabla^a\nabla^b\big(\Phi'(K) R_{iabj}\big).$$

The roadmap below uses only its schematic fourth-order part,

$$4\nabla^a\nabla^b\big(\Phi'(K) R_{iabj}\big).$$

The Berger theorems above are independent of a full PDE closure: they use the explicit homogeneous reduction. A complete DeTurck theorem would require a separate convention-by-convention comparison with the standard L^2-curvature-flow formula.

After a fourth-order DeTurck gauge, in the sense used for L^2-curvature type flows, the metric equation has the schematic leading form

$$\partial_t g = -4\varepsilon\Phi'(K)\Delta^2 g + \text{lower-order quasilinear terms},$$

provided the solution stays in a region

$$K \le K_P - \eta.$$

In such a region the coefficients of the equation are smooth. A full short-time existence theorem requires the corresponding DeTurck principal symbol and strong parabolicity verification. Positivity of $\Phi'(K)$ is the expected elliptic sign, but by itself it is not a substitute for the symbol calculation, since the linearisation also contains terms involving $\Phi''(K)$ and the background curvature.

For a metric variation $v = \partial_t g$, the Riemann tensor satisfies schematically

$$\partial_t \mathrm{Rm} = \nabla^2 v + \mathrm{Rm} * v.$$

Hence the wall part gives

$$\partial_t \mathrm{Rm} = -4\varepsilon\Phi'(K)\Delta^2 \mathrm{Rm} + \text{terms involving } \nabla\Phi', \nabla^2\Phi', \nabla^j \mathrm{Rm},$$

up to the index structure fixed by the Bianchi identities and by the DeTurck gauge.

Consequently

$$\partial_t K = 2\langle \mathrm{Rm}, \partial_t \mathrm{Rm}\rangle + \text{metric contraction terms}.$$

Thus the leading fourth-order contribution is not simply a scalar maximum-principle term

$$-4\varepsilon\Phi'(K)\Delta^2 K.$$

Rather, it contains

$$-8\varepsilon\Phi'(K)\langle \mathrm{Rm}, \Delta^2 \mathrm{Rm}\rangle$$

plus derivative terms coming from $\Phi'(K)$. After integration by parts these terms are expected to produce coercive fourth-order curvature dissipation, but pointwise they do not yield a classical maximum principle for K.

This distinction is essential. The fourth-order heat operator does not preserve positivity, and therefore no pointwise comparison argument of the form

$$\partial_t K \le F(K)$$

is available merely from the leading bi-Laplacian structure.

The singular energy nevertheless has a robust variational sign. Let

$$\mathcal{E}(t) = \int_M \Phi(K)\, d\mu.$$

Along the ungauged flow,

$$\frac{d}{dt}\mathcal{E} = \int_M \langle E_\Phi, -2\mathrm{Ric} - \varepsilon E_\Phi\rangle\, d\mu.$$

Hence

$$\frac{d}{dt}\mathcal{E} = -\varepsilon\|E_\Phi\|_{L^2}^2 - 2\int_M \langle E_\Phi, \mathrm{Ric}\rangle\, d\mu.$$

By Cauchy–Schwarz,

$$-2\int_M \langle E_\Phi, \mathrm{Ric}\rangle\, d\mu \le \frac{\varepsilon}{2}\|E_\Phi\|_{L^2}^2 + \frac{2}{\varepsilon}\|\mathrm{Ric}\|_{L^2}^2.$$

Therefore

$$\boxed{\frac{d}{dt}\mathcal{E} \le -\frac{\varepsilon}{2}\|E_\Phi\|_{L^2}^2 + \frac{2}{\varepsilon}\|\mathrm{Ric}\|_{L^2}^2.}$$

This is the correct energy-dissipation inequality. It shows that the wall term dissipates the singular energy up to the Ricci-flow error. It does not by itself prove pointwise wall separation.

A precise wall-separation statement requires more than the singular energy. It also requires spatial regularity and a lower local volume ratio. For example, suppose that on a time interval one has

$$\mathcal{E}(t) \le E_0, \qquad [K(\cdot,t)]_{C^\alpha} \le L,$$

and, for all sufficiently small r,

$$\mathrm{Vol}\, B(x,r) \ge v_0 r^3.$$

If

$$p > \frac{3}{\alpha},$$

then the singular energy bound forces a positive gap

$$\sup_M K(\cdot,t) \le K_P - \eta(E_0, L, v_0, p, \alpha).$$

Indeed, set

$$m = K_P - \sup_M K.$$

Choose x_0 with $K(x_0) \ge \sup_M K - m/2$. Then

$$K_P - K(x_0) \le \frac{3m}{2}.$$

By Hölder control,

$$K(x) \ge K(x_0) - L\, d(x, x_0)^\alpha.$$

Taking $r \asymp (m/L)^{1/\alpha}$, we have

$$K_P - K(x) \lesssim m$$

on $B(x_0, r)$. Therefore

$$\mathcal{E}(t) \ge \int_{B(x_0,r)} (K_P - K)^{-p}\, d\mu \gtrsim m^{-p} \operatorname{Vol} B(x_0, r) \gtrsim v_0 L^{-3/\alpha} m^{3/\alpha - p}.$$

When $p > 3/\alpha$, this lower bound diverges as $m \downarrow 0$, contradicting $\mathcal{E}(t) \le E_0$. Hence m is bounded below by a positive quantity depending only on E_0, L, v_0, p, α.

Thus the singular wall gives a plausible substitute for a maximum principle only in combination with spatial regularity and noncollapse-type local volume control. The missing analytic step is therefore clear:

energy dissipation + regularity/Sobolev control
+ local volume noncollapse $\Longrightarrow$ pointwise wall separation.

This correction is important: the wall energy alone cannot prove noncollapse. It can help prevent contact with $K = K_P$ only after enough local volume control has already been obtained by some independent mechanism.

In the homogeneous Berger reduction no such difficulty appears: all derivative terms vanish, and the wall reduces to the algebraic inward drift already computed. In the full PDE, however, the absence of a fourth-order maximum principle means that the Berger algebra must be supplemented by DeTurck parabolicity and a genuine energy-regularity argument.

19. Remaining noncollapse problem

The results above have a narrow conclusion. They identify a local high-curvature wall, rule out the natural additive Perelman-potential repairs tested on the Berger sector, and show that the unmodified Perelman entropy passes the protected positive-scalar Berger cone. They do not yet produce a full noncollapse functional for the inhomogeneous PDE.

A complete internal replacement for surgery would require a functional $\mathcal{N}$ with the following properties:

(i) a scale law compatible with the absolute curvature scale $K_P = \ell_P^{-4}$;
(ii) monotonicity along the actual forced DeTurck system;
(iii) a coercive implication from a lower bound for $\mathcal{N}$ to local volume noncollapse under a curvature bound;
(iv) compatibility with bounded-curvature Hopf-fibre collapse, which is invisible to the wall $K \nearrow K_P$.

The preceding results indicate that such an $\mathcal{N}$, if it exists for the full PDE, cannot be obtained merely by adding a K-potential to Perelman's entropy and using the same K-gradient to force the metric. The Berger calculation points instead to a two-mechanism design: the curvature wall controls high curvature, while an independent non-additive detector, beginning

with the unmodified Perelman entropy in the homogeneous sector, supplies noncollapse information. Other structurally different candidates include modified reduced volume, Harnack-type quantities for the coupled system, spacetime action functionals, and heat-kernel detectors.

Open Problem 1 (PDE propagation of the non-additive detector)**.** *The protected Berger cone is passed by the unmodified Perelman entropy. The open problem is to prove an analogue for the full inhomogeneous wall flow: either a monotonicity formula or a controlled propagation estimate for the unmodified entropy, sufficient to retain fixed-scale volume noncollapse under $K \le r^{-4} < K_P$, and compatible with a DeTurck continuation theorem.*

20. Conclusion

This paper should be read as a mathematical side branch of the compact S^3 programme, not as a required physical extension of the operational Planck-boundary interpretation. It studies modified Ricci-type flows as autonomous geometric dynamics and asks which variational curvature barriers suppress supercritical curvature growth. The parameter $K_P = \ell_P^{-4}$ serves here as a prescribed curvature-wall scale; it is not being promoted to a new physical forcing law.

A constant cosmological term fails the high-curvature test because its curvature contribution is quadratic in $|\mathrm{Rm}|$, while the Ricci reaction is cubic. The variational wall

$$-\varepsilon \operatorname{grad}_{L^2(d\mu)} \int_M \frac{(|\mathrm{Rm}|^2+\delta)^q}{(K_P - |\mathrm{Rm}|^2)^p}\, d\mu$$

is the corresponding local high-curvature replacement. In dimension three it acts on the full curvature invariant

$$K = |\mathrm{Rm}|^2 = 4|\mathrm{Ric}_0|^2 + \frac{1}{3}R^2,$$

and its leading conformal drift has the correct sign and degree.

This is not a noncollapse theorem. The Hopf-fibre degeneration in the Berger family gives bounded-curvature collapse, so a K-wall alone cannot replace surgery. The exact Berger anisotropy equation shows the same point dynamically: in the cone $u \ge 4/11$ the round ratio $u = 1$ is attracting, whereas in the lower Hopf sector the wall can push u downward when

$$\frac{2\varepsilon\Phi'(K)}{x}(4 - 11u) > 1.$$

Volume normalization does not alter this anisotropy equation, and the Hopf limit is an S^2-quotient collapse rather than a high-curvature limit. The heat-kernel scale detector $\mathcal{H}_r$ diverges along fixed-base Hopf collapse by the zero-mode lower bound, which identifies a genuinely non-additive direction for future noncollapse tests. The natural attempt to turn the wall into an additive Perelman-type noncollapse entropy fails: $R + \varepsilon\Phi(K)$ is not monotone even in the homogeneous Berger sector, and additive matching K-potential repairs fail for every sign and every τ-weight in the pure quartic test. The unmodified Perelman entropy behaves differently: on the protected positive-scalar Berger cone $4/11 \le u < 4$, its reduced derivative along the wall flow is nonnegative. This gives a positive non-additive two-mechanism Berger result, not a full PDE theorem.

The final conclusion is therefore precise:

the curvature wall is a valid local high-curvature regulator;

matching additive K- and Planck-wall potentials do not supply noncollapse;

the unmodified Perelman entropy supplies a positive Berger two-mechanism test;

a complete surgery replacement still requires PDE propagation of this detector.

Thus the contribution of this paper is a mathematical one. It develops a curvature-wall mechanism for modified Ricci-type flows and tests it against Berger collapse, additive entropy repairs, heat-kernel noncollapse indicators, and the unmodified Perelman entropy. It should be read as an optional mathematical refinement, not as a required physical extension of the operational Planck-boundary interpretation.

Any future candidate, including the unmodified entropy, must pass from the Berger reduction to an inhomogeneous DeTurck continuation theorem.

Appendix A. Algebraic checks

A.1. **Berger curvature.** With $x = a^2$, $z = c^2$, and $u = z/x$,

$$|\mathrm{Rm}|^2 = \frac{4(16a^4 - 24a^2c^2 + 11c^4)}{a^8} = \frac{4(16x^2 - 24xz + 11z^2)}{x^4} = \frac{4(16 - 24u + 11u^2)}{x^2}.$$

At $u = 1$,

$$|\mathrm{Rm}|^2 = \frac{12}{x^2}.$$

At fibre collapse $u \to 0$ with x fixed,

$$|\mathrm{Rm}|^2 \to \frac{64}{x^2} = \frac{64}{a^4}.$$

At base collapse $x \to 0$ with u fixed,

$$|\mathrm{Rm}|^2 \to \infty$$

because

$$11u^2 - 24u + 16 > 0$$

for all real u, its discriminant being

$$(-24)^2 - 4 \cdot 11 \cdot 16 = -128 < 0.$$

A.2. **Scale exponent.** For a length scale L,

$$\int K^q \, d\mu \sim L^{-4q} L^3 = L^{3-4q}.$$

The critical exponent is therefore

$$3 - 4q = 0 \quad \Longleftrightarrow \quad q = \frac{3}{4}.$$

A.3. **Power law.** For $\Phi(K) = K^q$,

$$\Phi'(K) = qK^{q-1},$$

and hence

$$4K\Phi'(K) - 3\Phi(K) = 4qK^q - 3K^q = (4q - 3)K^q.$$

A.4. **Barrier law.** For

$$\Phi(K) = \frac{(K + \delta)^q}{(K_P - K)^p},$$

one has

$$\frac{\Phi'(K)}{\Phi(K)} = \frac{q}{K + \delta} + \frac{p}{K_P - K}.$$

Thus

$$4K\Phi'(K) - 3\Phi(K) = \Phi(K) \left(\frac{4qK}{K + \delta} + \frac{4pK}{K_P - K} - 3 \right).$$

A.5. **Sturm check for the Berger falsification.** The polynomial

$$c_2(u) = 1331u^4 - 3696u^3 + 3648u^2 - 1536u + 256$$

has no positive root. One Sturm sequence begins with

$$c_2, \quad c_2' = 5324u^3 - 11088u^2 + 7296u - 1536,$$

and continues, up to positive scalar multiples, by

$$1104u^2 - 1256u + 116, \qquad 48384 - 48661u, \qquad 217220096.$$

The sign variation count at $u = 0^+$ equals the sign variation count at $u = +\infty$, so the number of positive roots is zero. Since $c_2(0) = 256 > 0$, one has $c_2(u) > 0$ for all $u > 0$.

Appendix B. Reference facts used

The use of L^2-curvature-gradient flows as fourth-order degenerate geometric flows with DeTurck gauge fixing follows the standard theory of the gradient flow of $\int_M |\mathrm{Rm}|^2$, especially Streets' DeTurck treatment of the L^2-curvature flow. The heat-kernel coercivity discussion uses the standard local comparison theory under a Ricci lower bound, in the spirit of Cheeger–Yau, Li–Yau, and Davies. The tensorial formulas in the analytic roadmap are used only at the schematic principal-structure level; a complete PDE theorem would require a separate convention-by-convention verification against that standard L^2-flow formula. The restriction on second-order natural divergence-free tensors is the Lovelock-type obstruction: second-order metric-natural tensors have very limited algebraic freedom, and in low dimension do not provide a new high-curvature damping mechanism.

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